

Preserving the Optimal-Flexibility Condition of a Pulsed-Jet Nozzle at Millimetre Scale

Maya Angia

Bhamla Lab, Georgia Institute of Technology

Abstract. A compliant nozzle can raise the hydrodynamic impulse of a pulsed jet by up to roughly 400% relative to a rigid nozzle, but only when its elastic response is tuned to the jet: the governing condition is a dimensionless wave speed $\hat{c} = cT_{\text{acc}}/L \simeq 3$, established at centimetre scale by Choi and Park. Whether that condition survives a tenfold reduction in nozzle diameter is an open question, and answering it experimentally requires a nozzle whose length and wall thickness are chosen so that $\hat{c}$ is held fixed rather than allowed to drift with the geometry. This note derives the two design relations needed for that reduction. First, under strict geometric similarity at 1/10 scale ($D = 1.5$ mm), the wall thickness that preserves $\hat{c}$ is $h = 25$ μm and the nozzle length is unchanged from the full-scale value. Second, because a 25 μm elastomer wall is at or beyond the limit of reliable casting, any thicker fabricable wall must be compensated by lengthening the nozzle as $L \propto \sqrt{h}$. Length values are tabulated for wall thicknesses of 25–250 μm against target $\hat{c}$ from 2.2 to 13.0, and the operating point nearest the published optimum is identified as $L \approx 21.7$ mm. The tabulation converts an otherwise open fabrication choice into a bounded one and defines the geometry for a millimetre-scale squid-inspired propulsor.

Keywords. pulsed jet; flexible nozzle; fluid–structure interaction; dimensionless wave speed; dynamic similarity; bio-inspired propulsion; cephalopod locomotion

1 Motivation

Cephalopods and salps propel themselves by ejecting discrete slugs of fluid through a compliant funnel, and the compliance is not incidental. Choi and Park [1] showed experimentally that a nozzle of correctly tuned flexibility stores elastic energy during jet acceleration and returns it to the jet as the nozzle contracts, raising hydrodynamic impulse by approximately 400% and entrained volume by approximately 220% over a rigid nozzle of the same bore. The same study reports that a flexible nozzle used as a thruster on a small underwater vehicle increased translation speed by roughly 60% at the optimal condition.

The enhancement is not monotonic in compliance. It peaks when the travel time of the elastic surface wave along the nozzle matches the jet acceleration time. That matching condition is expressed as a single dimensionless group and its optimum was located at $\hat{c} \simeq 3.0$ for both continuous [2] and pulsed [1] jets, using nozzles of inner diameter $D = 15$ mm.

Whether the mechanism persists at the scale of a centimetre-class swimming robot is not established. Reducing D by a factor of ten reduces the fluid inertia term in the wave speed, and if wall thickness and nozzle length are carried over unchanged the operating point moves off the optimum, which would confound a negative result with a design error. The purpose of this note is to fix the geometry so that $\hat{c}$ is a controlled variable in the scaled experiment rather than a consequence of whatever wall the casting process happens to yield.

2 Governing groups

Following Choi and Park [1,2], the elastic surface wave on a thin-walled cylindrical nozzle propagates at

$$c = \sqrt{\frac{Eh}{\rho_f D}}, \quad (1)$$

where E is the Young's modulus of the nozzle wall, h the wall thickness, ρ_f the fluid density and D the undeformed inner diameter. The product Eh is the structural stiffness of the wall and is the quantity that is physically measured, since E and h are not independently controlled in a cast elastomer.

Normalising c by the characteristic velocity L/T_{acc} , where L is the nozzle length and T_{acc} the jet acceleration time, gives the dimensionless wave speed

$$\hat{c} = \frac{cT_{\text{acc}}}{L} = \sqrt{\frac{EhT_{\text{acc}}^2}{\rho_f DL^2}}. \quad (2)$$

Equation (2) can be written as $\hat{c} = \sqrt{\Pi_0^2 \Pi_1 / 2}$ in terms of the effective acceleration time $\Pi_0 = v_{m,r} T_{\text{acc}} / L$ and the effective nozzle stiffness $\Pi_1 = Eh / (\rho_f v_{m,r}^2 R)$, with $v_{m,r}$ the peak jet-exit velocity for the rigid reference nozzle and $R = D/2$ [1]. Impulse and entrainment are both maximised at $\hat{c} \simeq 3.0$. Above that value the wall is too stiff and contracts before the jet reaches peak velocity, releasing the stored energy too early; below it, contraction is delayed past the end of the pulse and the transfer is incomplete.

The reference condition used throughout this note is the elastic-wave travel distance during one acceleration interval,

$$\Lambda \equiv c T_{\text{acc}} = 64.95 \text{ mm}, \quad (3)$$

evaluated for $Eh = 14.4 \text{ N m}^{-1}$, $D = 15 \text{ mm}$, $\rho_f = 1000 \text{ kg m}^{-3}$ and $T_{\text{acc}} \approx 66 \text{ ms}$, which lies inside the range $T_{\text{acc}} = 0.05\text{--}0.16 \text{ s}$ reported in [1]. With Λ fixed, equation (2) reduces to the hyperbola

$$\hat{c} = \frac{\Lambda}{L}, \quad (4)$$

so that specifying a target $\hat{c}$ specifies a nozzle length directly. The published optimum $\hat{c} = 3.0$ corresponds to $L = 21.65 \text{ mm}$.

3 The similarity wall thickness at one-tenth scale

Holding E , ρ_f and T_{acc} fixed, equations (1) and (2) give

$$\hat{c} \propto \frac{1}{L} \sqrt{\frac{h}{D}}, \quad (5)$$

from which two statements follow immediately.

3.1 Geometric similarity preserves the operating point

If the nozzle is scaled geometrically, so that h/D is held constant, then $\hat{c} \propto 1/L$ with the same constant Λ , and the nozzle length required to hit a given $\hat{c}$ is *unchanged* by the reduction in diameter. This is a convenient result and a slightly counter-intuitive one: the scaled nozzle is not shorter.

The full-scale reference wall is $h = 250 \mu\text{m}$ at $D = 15 \text{ mm}$, giving $h/D = 1/60$. At $D = 1.5 \text{ mm}$ the geometrically similar wall is therefore

$$h_{\text{ref}} = \frac{D}{60} = 25 \mu\text{m}. \quad (6)$$

3.2 A thicker wall must be paid for in length

A 25 μm elastomer wall cast as a free-standing tube is at the edge of what can be produced repeatably, and a wall that tears during the first pulse yields no data at all. For any fabricable thickness $h > h_{\text{ref}}$, equation (5) requires

$$L(h) = L_{\text{ref}} \sqrt{\frac{h}{h_{\text{ref}}}}, \quad h_{\text{ref}} = 25 \mu\text{m}, \quad (7)$$

where L_{ref} is the length that delivers the target $\hat{c}$ under geometric similarity, i.e. $L_{\text{ref}} = \Lambda/\hat{c}$. Thickening the wall by a factor of ten therefore costs a factor of $\sqrt{10} \approx 3.16$ in length. Fabrication ease and compactness are in direct tension, and equation (7) states the exchange rate.

4 Tabulated geometry

Table 1 evaluates equations (4) and (7) across the fabricable range. Columns are wall thicknesses attainable by successive dip-coating or thin-film casting passes; rows are target values of $\hat{c}$ spanning the published range 1.1–16.0. Every cell is a candidate nozzle geometry that sits at the same point on the impulse curve.

Table 1. Required nozzle length L (mm) at $D = 1.5$ mm as a function of wall thickness h and target dimensionless wave speed $\hat{c}$, from equations (4) and (7). L_{ref} is the length under geometric similarity ($h = 25 \mu\text{m}$). The row nearest the published optimum $\hat{c} \simeq 3.0$ is set in bold.

L_{ref} (mm)	$\hat{c}$ ($= \Lambda/L_{\text{ref}}$)	Required L (mm) for wall thickness h (μm)				
		25	50	100	200	250
5	12.99	5	7.07	10	14.14	15.81
10	6.50	10	14.14	20	28.28	31.62
15	4.33	15	21.21	30	42.43	47.43
20	3.25	20	28.28	40	56.57	63.25
25	2.60	25	35.36	50	70.71	79.06
30	2.17	30	42.43	60	84.85	94.87

Three observations govern the build.

1. **The optimum is at $L_{\text{ref}} = 21.65$ mm.** The tabulated row $L_{\text{ref}} = 20$ mm gives $\hat{c} = 3.25$, an 8% overshoot. Since impulse falls off sharply for $\hat{c} > 3$ and only gradually below it [1], rounding the target length *up* rather than down is the safer error.
2. **Only the leftmost two columns are compact.** At $h = 250 \mu\text{m}$ the nozzle must be 63 mm long to stay on the optimum, which is longer than the intended vehicle. Wall thickness is therefore the binding constraint on whether a scaled propulsor is buildable at all, not stiffness selection.
3. **The whole table is one curve.** Every entry has the same $\hat{c}$ within its row, so a disagreement between two rows in measured impulse is evidence about $\hat{c}$, whereas a disagreement *within* a row is evidence that some other group, most plausibly the Reynolds number, has not been held.

5 Selected operating point and expected checks

The build target is $D = 1.5$ mm, $h = 50 \mu\text{m}$, $L = 28.3$ mm, giving $\hat{c} = 3.25$. This sits one fabrication step away from the similarity wall, keeps overall length under 30 mm, and places the geometry within 10% of the

published optimum.

At this scale the jet Reynolds number $Re = v_{m,r}D/\nu$ falls by the same factor of ten as the diameter, so the scaled experiment does not preserve Re . That is expected and is precisely the residual question: the elastic energy-return mechanism is argued in [1] to be governed by the fluid–structure timescale ratio rather than by the state of the shear layer, and the scaled build is a direct test of that claim. Two outcomes are informative. If impulse enhancement is retained near $\hat{c} \simeq 3$, the mechanism is confirmed as Re -insensitive over a decade and the design rule transfers to small robots. If enhancement collapses while $\hat{c}$ is held fixed, the mechanism depends on something the three groups do not capture, which is the more interesting result and the one worth reporting.

6 Limitations

The reference constant Λ in equation (3) is reconstructed from the stiffness, diameter and acceleration-time ranges reported in [1] rather than measured on the present apparatus; it should be re-derived from a direct measurement of Eh and T_{acc} once the scaled jet generator is characterised. Equation (1) assumes a thin, uniform, linearly elastic wall and no axial tension, none of which is exactly true of a cast elastomer with a thickened support collar. Wall thickness in a dip-cast tube varies along the axis, so h in Table 1 should be read as a length-averaged value, consistent with the definition used in [1]. Finally, all lengths assume the elastomer modulus is unchanged by the reduction in wall thickness, which will fail if the cast film is thin enough for surface effects or incomplete cross-linking to alter E .

References

- Choi, D. and Park, H. (2024). Mechanism of enhanced impulse and entrainment of a pulsed jet through a flexible nozzle. *Journal of Fluid Mechanics*, 996, A6. doi:10.1017/jfm.2024.720. Open access under CC BY 4.0.
- Choi, D. and Park, H. (2022). Optimal flexibility of a flexible nozzle for maximising the thrust of a jet. *Journal of Fluid Mechanics*, 949, A39. doi:10.1017/jfm.2022.771.
- Gharib, M., Rambod, E. and Shariff, K. (1998). A universal time scale for vortex ring formation. *Journal of Fluid Mechanics*, 360, 121–140. doi:10.1017/S0022112097008410.
- Krueger, P. S. and Gharib, M. (2003). The significance of vortex ring formation to the impulse and thrust of a starting jet. *Physics of Fluids*, 15(5), 1271–1281. doi:10.1063/1.1564600.
- Staaf, D. J., Gilly, W. F. and Denny, M. W. (2014). Aperture effects in squid jet propulsion. *Journal of Experimental Biology*, 217(9), 1588–1600. doi:10.1242/jeb.082271.
- Bujard, T., Giorgio-Serchi, F. and Weymouth, G. D. (2021). A resonant squid-inspired robot unlocks biological propulsive efficiency. *Science Robotics*, 6(50), eabd2971. doi:10.1126/scirobotics.abd2971.

Document note. This note formalises scaling calculations produced during a millimetre-scale flexible-nozzle build in the Bhamla Lab. The dimensionless groups, the wave-speed relation and the optimum $\hat{c} \simeq 3.0$ are results of Choi and Park [1,2] and are cited as such; the similarity wall thickness of equation (6), the length–thickness exchange of equation (7) and Table 1 are the contribution recorded here. The work is unpublished and is not part of reference [1].